TDWG Standard · GBIF & OBIS · Open Biodiversity Data

Darwin Core (DwC), Data Publishing, Citation and IPT

Mobilizing, standardizing, and crediting global biodiversity records across the Open Science research lifecycle.

Begin Lecture →

Open Science Course

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The Problem: The Biodiversity Tower of Babel

Before standardization, natural history collections and field scientists recorded data in disparate silos:

🏷️ Taxonomy Chaos

Synonyms, typos, regional vernacular names, and conflicting authorities without persistent identifiers.

🗺️ Spatial Ambiguity

"Behind the old mill" vs degrees/minutes/seconds vs local UTM grids; lack of coordinate uncertainty metrics.

📅 Temporal Inconsistency

`04/05/06` (May 4th or April 5th?), arbitrary date intervals, time zones missing.

🎯 **The Consequence:** Global synthesis (climate change range shifts, extinction tracking, invasive species modeling) was virtually impossible without manual data wrangling.

What is Darwin Core (DwC)?

Darwin Core (DwC) is the international standard maintained by **TDWG (Biodiversity Information Standards)**.

- **A Glossary of Terms:** A stable vocabulary with precise URI definitions (e.g. `dwc:occurrenceID`, `dwc:scientificName`).
- **Pragmatic & Flexible:** Storage-independent (flat CSV, relational SQL, JSON-LD, RDF XML).
- **The Lingua Franca of Global Portals:** The foundational exchange format for **GBIF**, **OBIS**, and national biodiversity nodes.

🔗 **Essential Reference Guide:**
Consult the official Darwin Core Quick Reference Guide (dwc.tdwg.org) for every term definition, namespace, and normative examples.

Official TDWG Identity tdwg.org ↗

Standard:	**Biodiversity Information Standards (TDWG)**
Ratified:	October 2009 (Living standard)
Guide URL:	dwc.tdwg.org
Namespace:	http://rs.tdwg.org/dwc/terms/
Quick List:	dwc.tdwg.org/list/

"Darwin Core is primarily based on taxa, their occurrence in nature as documented by observations, specimens, samples, and related information."

What is GBIF?

The **Global Biodiversity Information Facility** (GBIF.org) is an international network and research infrastructure funded by governments worldwide.

The World's Largest Biodiversity Aggregator

GBIF provides free and open access to biodiversity data globally, mobilizing evidence from museum specimens, ecological surveys, eDNA, and citizen science (e.g. iNaturalist, eBird).

Global Open Mandate: Standardized open licenses (CC0, CC-BY, CC-BY-NC) and citable DOIs for all datasets and search queries.

Powered by Darwin Core: Universal lingua franca harmonizing disparate records into a single queryable global index.

Conservation & Policy: Essential evidence for IUCN Red List assessments, IPBES reports, and biodiversity targets.

2.9+ Billion
Species Occurrence Records
Museum specimens, citizen science, eDNA, sensors

2,500+
Publishing Institutions
Universities, museums, government agencies, NGOs

100+
Voting Participants & Nodes
Country delegations and international organizations

How GBIF Operates: Nodes & Governance

GBIF is **not an uncurated upload bucket** — it is a **federated, endorsed network**:



Country / Participant Nodes

Designated national teams (e.g. GBIF Norway, GBIF Spain) serving as the official bridge between local institutions and the global infrastructure.

Institutional Endorsement

Strict vetting: Node managers must review and approve every publishing organization before its datasets can be registered on GBIF.

Federated Publishing (IPT)

Organizations keep ownership of their data. The node hosts or assists with the **IPT**, which exports Darwin Core Archives for GBIF to harvest.

GBIF Dataset Classes Overview

GBIF standardizes biodiversity publishing into **4 distinct dataset classes** (gbif.org/dataset-classes):

1. Occurrence

`Core: dwc:Occurrence`

Evidence of a species at a given time and place (museum specimens, citizen science observations, camera traps).

`Most common on GBIF`

2. Sampling-Event

`Core: dwc:Event + Occ Ext`

Standardized environmental monitoring, surveys, transects with quantifiable effort and non-detection (absence) data.

`Ecological time-series`

3. Checklist

`Core: dwc:Taxon`

Catalogues of taxa, national inventories, red lists, and taxonomic classifications without specific coordinate events.

`Taxonomic backbone`

4. Metadata-only

`Core: None (EML XML)`

Describing undigitized physical collections, historical literature, or projects awaiting electronic mobilization.

`Collection discovery`

GBIF Recommendations: Mandatory vs. Recommended Terms

Explore terms per dataset class based on gbif.org/dataset-classes and dwc.tdwg.org:

Occurrence | Sampling-Event | Checklist | Metadata-Only

All | ● Mandatory Only

Occurrence Datasets (Core: dwc:Occurrence)

GBIF Class Docs

Evidence of the presence, abundance, or absence of a biological organism at a specific location and date.

Darwin Core Term	GBIF Status	Definition & Purpose	Example Value
dwc:occurrenceID Occurrence	● Mandatory	Globally unique, persistent identifier for the occurrence record. ↳ Primary key; GBIF indexing fails without it	urn:uuid:8b3e8e2b-2e92-498c-851c-a1288c8eb25a
dwc:basisOfRecord Occurrence	● Mandatory	Controlled vocabulary denoting the specific nature of the record. ↳ Required controlled vocabulary for filtering	HumanObservation, PreservedSpecimen, MaterialSample

The Anatomy of a Darwin Core Record

A practical example of how a biological observation maps directly to standardized DwC terms:

```
{
  "dwc:occurrenceID": "urn:uuid:8b3e8e2b-2e92-498c-851c-a1288c8eb25a",
  "dwc:basisOfRecord": "HumanObservation",
  "dwc:scientificName": "Bombus terrestris (Linnaeus, 1758)",
  "dwc:kingdom": "Animalia",
  "dwc:taxonRank": "species",
  "dwc:decimalLatitude": 50.0755,
  "dwc:decimalLongitude": 14.4378,
  "dwc:geodeticDatum": "WGS84",
  "dwc:coordinateUncertaintyInMeters": 15,
  "dwc:eventDate": "2024-06-12T10:15:00Z",
  "dwc:recordedBy": "Michal Torma",
  "dcterms:license": "http://creativecommons.org/publicdomain/zero/1.0/"
}
```

Mandatory Core Terms

- `occurrenceID`
- `basisOfRecord`
- `scientificName`
- `eventDate`

Strongly Recommended

- `decimalLatitude`
- `decimalLongitude`
- `geodeticDatum`
- `coordinateUncertaintyInMeters`

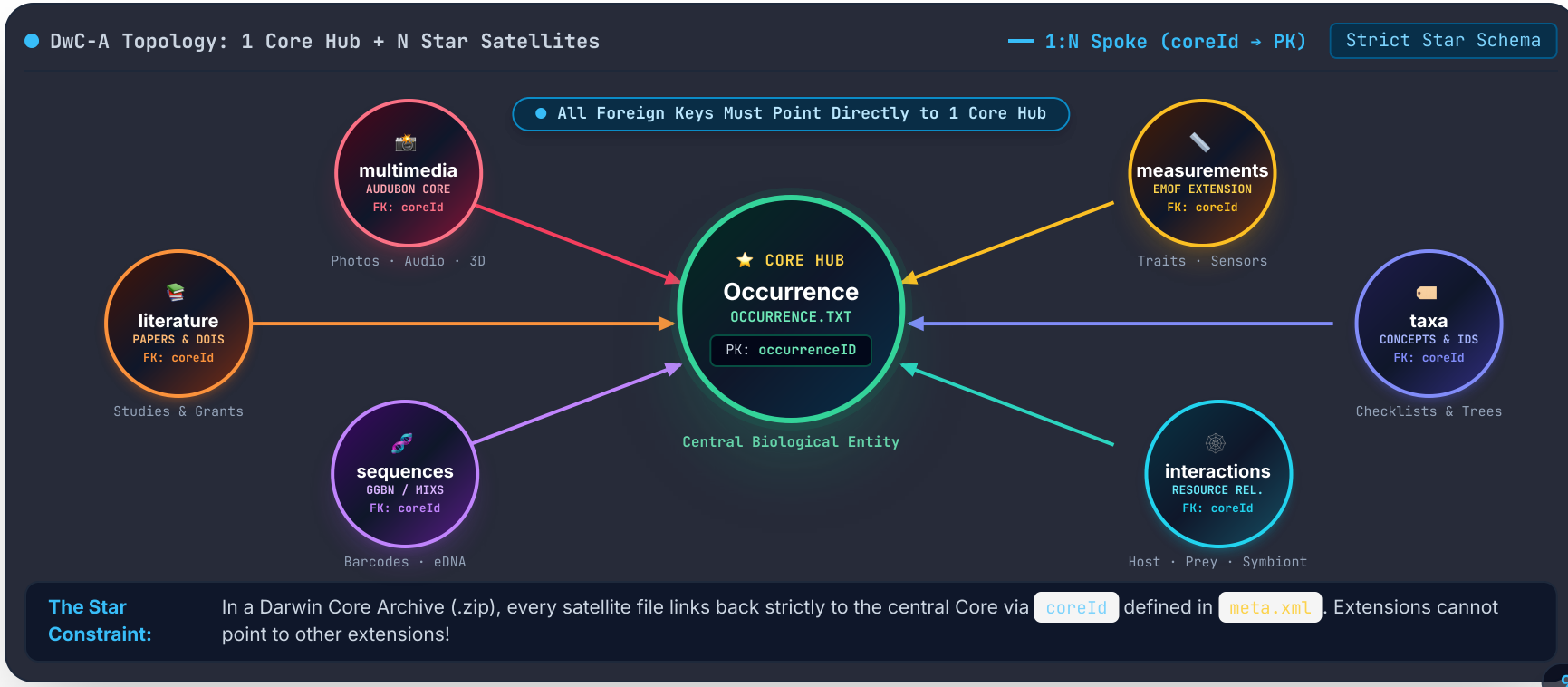
Mandatory Open License

Machine-readable license URL is required for GBIF publication:

- `CC0 1.0`
- `CC-BY 4.0`
- `CC-BY-NC 4.0`

The Darwin Core Star Schema (DwC-A)

A single flat table cannot cleanly model **one-to-many (1:N)** biological relationships. A **Darwin Core Archive (DwC-A)** solves this using a **Star Schema**:



GBIF Extensions in Practice

GBIF maintains a community-curated Extension Registry to extend Core records with specialized data types:

📸 Simple Multimedia (Audubon Core)

Media

Attaches photos, field audio recordings, or 3D specimen scans to an occurrence record.

Terms: `identifier`, `format`, `license`

iNaturalist · Herbarium scans

🔪 Extended MeasurementOrFact (eMoF)

Traits & Sensors

Records quantitative traits (body length, mass) and environmental metrics (water temp, depth).

Terms: `measurementType`, `Value`, `Unit`

OBIS · Trait databases

🧬 DNA Derived Data (GGBN / MIxS)

eDNA & Sequences

Links DNA barcoding, metabarcoding, and environmental DNA (eDNA) detections to vouchers.

Terms: `target_gene`, `pcr_primer`, `accession`

NCBI · BOLD · ENA

🔗 Resource Relationship

Interactions

Documents ecological interactions between occurrences (host-parasite, pollinator, prey).

Terms: `relationshipOfResource`, `relatedResourceID`

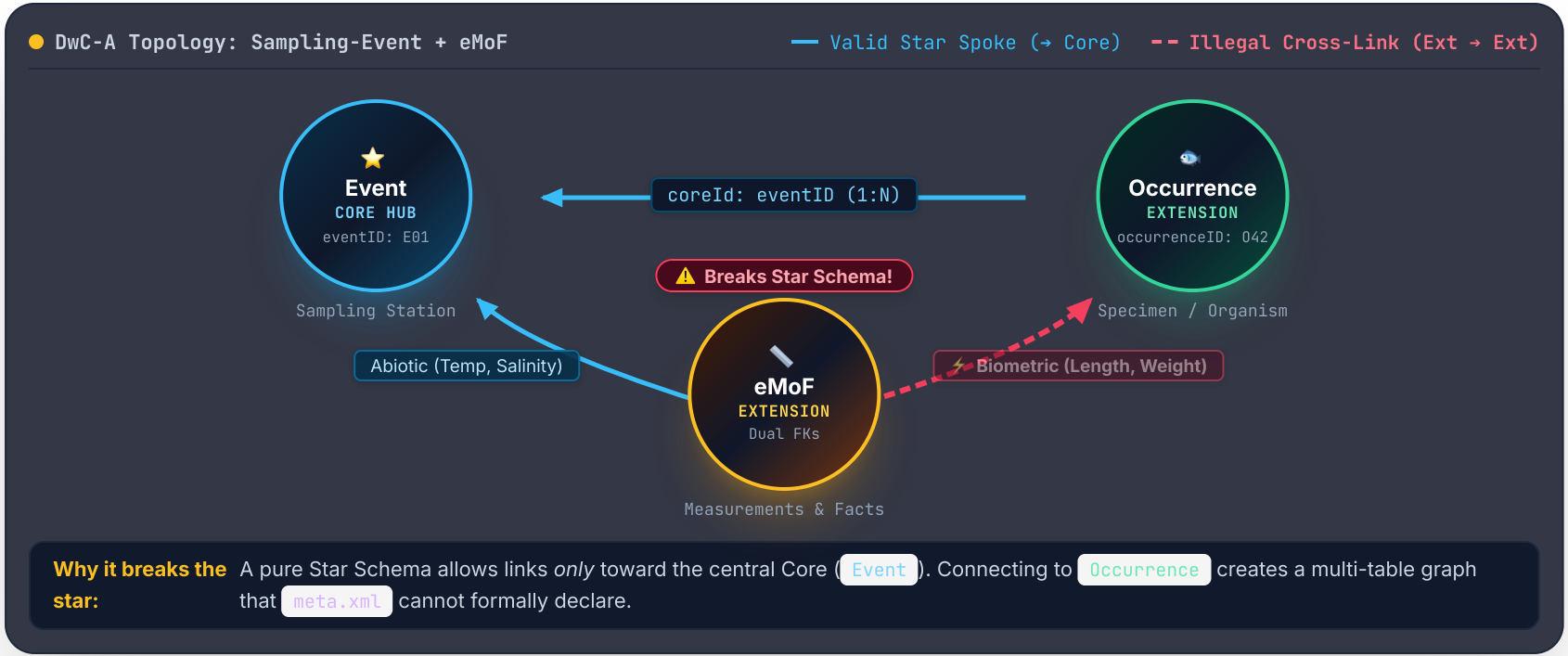
GloBI · Food webs

💡 Each extension file links back to `coreId` in `occurrence.txt` via schema defined in `meta.xml`

Explore Registry ↗

Special Case: ExtendedMeasurementOrFact (eMoF)

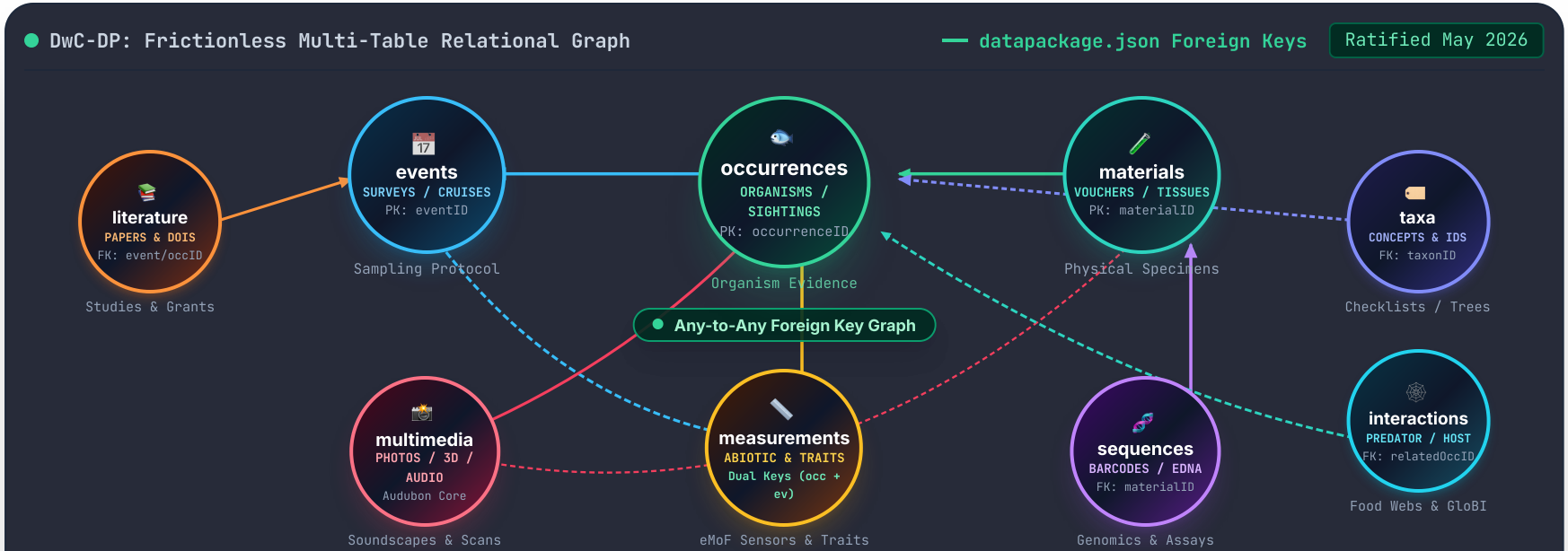
In Sampling-Event datasets, measurements target **both** the Event (abiotic) and Occurrence (biometric traits), breaking the pure Star Schema:



Why it breaks the star: A pure Star Schema allows links *only* toward the central Core (`Event`). Connecting to `Occurrence` creates a multi-table graph that `meta.xml` cannot formally declare.

The Solution: Darwin Core Data Package (DwC-DP)

Officially ratified by **TDWG on May 26, 2026**: Replacing the rigid Star Schema with Frictionless Data Packages and multi-table relational graphs.





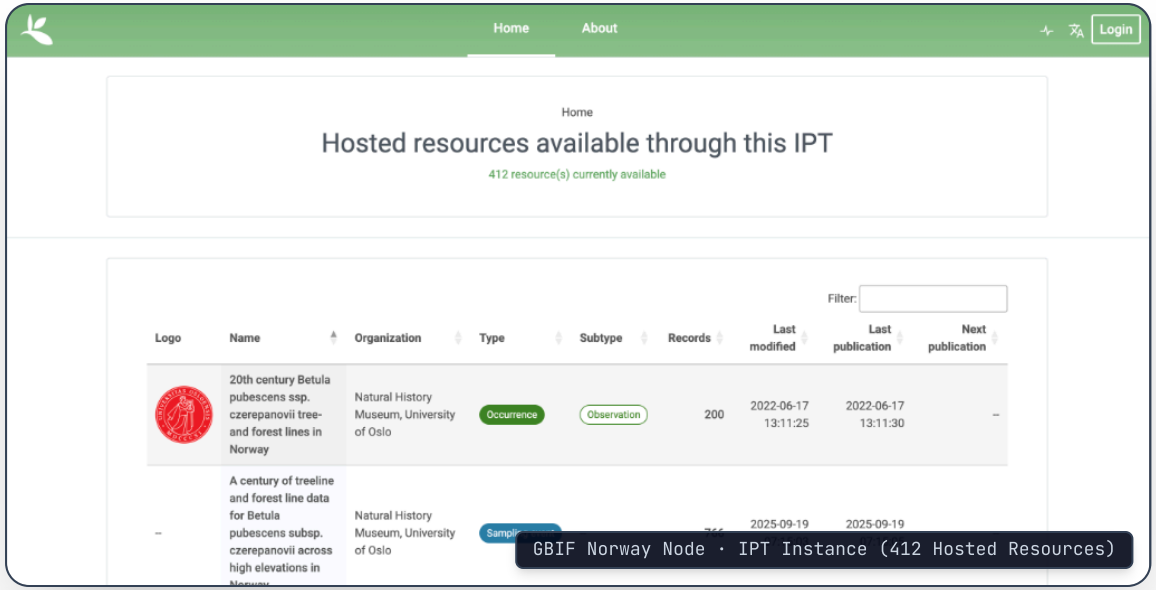
"That looks complicated..."

...I hope there are tools to help me!

Don't worry — you don't have to build these archives or schemas by hand!

Meet the Integrated Publishing Toolkit (IPT)

GBIF's open-source web application designed to manage, document, and publish biodiversity datasets without writing raw XML or JSON.



GBIF Norway Node · IPT Instance (412 Hosted Resources)

What is the IPT?

GBIF's free open-source publishing tool (Java web app), typically hosted by **national participant nodes** or institutions.

Automated Packaging

Upload CSV/Excel tables or connect databases. IPT maps your fields to **DwC terms** and packages valid archives automatically.

Direct Node Registration

Registers datasets directly into the **GBIF Registry** with persistent DOIs and scheduled crawler sync.

IPT Workflow · Step 1: Source Data

Connecting raw observation files or live database queries to the IPT resource.



IPT Source Data Interface: File or SQL database table connection

📁 Supported Formats

Delimited text files (**CSV, TSV**), Microsoft Excel spreadsheets, or direct live **SQL database connections** (PostgreSQL, MySQL, Oracle).

⚡ Scalability

Handles small field survey lists up to millions of rows (as seen here with **337 MB / 76,882 rows** in an occurrence source).

🔒 Preserves Original Data

Your original column headers and internal IDs remain untouched — mapping to Darwin Core happens in the next step.

IPT Workflow · Step 2: Darwin Core Mappings

Translating local column headers into standardized Darwin Core terms and schemas.



Core & Extension Mappings: Mapping 38 Darwin Core terms

🎯 Core vs. Extension Selection

Designate your main table (e.g. **Darwin Core Occurrence** or **Event**), and attach any number of registered extensions.

🪄 Smart Auto-Mapping

The IPT inspects column names and automatically matches standard terms (e.g. `lat` → `decimalLatitude`).

🔍 Value Translation & Vocabularies

Map internal codes (e.g., `OBS`) to controlled TDWG vocabularies (`HumanObservation`) directly in the UI.

IPT Workflow · Step 3: Resource Metadata

Describing the Who, What, Where, When, and Why using Ecological Metadata Language (EML).



IPT Metadata Status: Validated EML document ready for publication

📋 Comprehensive EML Profile

Fills out dataset title, abstract, creators, contacts, geographic bounding boxes, taxonomic scope, and temporal ranges.

✅ Real-Time Validation Pill

The green `Valid` badge confirms all mandatory GBIF and DataCite citation fields are complete.

⚖️ Machine-Readable Licensing

Select standard open licenses (**CC0 1.0, CC-BY 4.0, or CC-BY-NC 4.0**) to legally govern reuse and citation credit.

IPT Workflow · Step 4: Visibility & Registration

Governing dataset access: from private local staging to endorsed public discovery.



Public Resource: Registered with GBIF and discoverable worldwide

🔒 Private (Draft Mode)

Accessible only to authenticated resource managers. Ideal for internal verification, data cleaning, and co-author review.

🔓 Public (Direct Link)

Anyone with the IPT resource URL can download the generated DwC-A archive and read metadata, but not yet indexed on GBIF.org.

🌐 Registered (GBIF Network)

Endorsed by a **Participant Node** and registered in the GBIF Registry. Automatically indexed and citable worldwide.

IPT Workflow · Step 5: Publication & Versioning

Publishing major/minor versions, minting DOIs, and scheduling automated background sync.



Version Control: Live version 1.457 vs. scheduled daily publication 1.458

⏱️ Automated Re-publishing

As shown here (`Enabled: Daily`), databases update records automatically every day without human intervention!

🏷️ Semantic Versioning & DOIs

Each major release receives an immutable version tag (`1.457 → 1.458`) and DOI tracking, keeping scientific citations fully reproducible.

📦 Instant Package Generation

Compiles the ZIP archive containing `meta.xml`, `eml.xml`, and standardized DwC tables for GBIF harvester crawling.

IPT Workflow · Step 6: GBIF Registration

Attaching the dataset to an endorsed publishing institution and minting global registry keys.



GBIF Registration: Linked UUID key under Norwegian Environment Agency

🏛️ Publishing Organization

Links the dataset to an endorsed publisher (e.g. *Norwegian Environment Agency*) approved by a GBIF Participant Node.

🔑 GBIF Dataset UUID

Assigns an immutable global UUID (`56fb17ca-adb2...`) in the global GBIF Registry.

🤖 Automated Harvester Hook

Tells the central GBIF crawler where to retrieve updates whenever a new version is published in this IPT.

IPT Workflow · Step 7: Thematic Networks

Connecting datasets to specialized global initiatives (e.g. OBIS, VertNet, national consortia).



GBIF Networks: Grouping datasets into collaborative thematic umbrellas

Thematic Aggregation

Allows associating the resource with specialized networks like **OBIS** (marine), **VertNet** (vertebrates), or national monitoring programs.

Cross-Dataset Portals

Enables custom filtering and unified analytics across multi-institutional consortia publishing through different IPTs.

Optional Participation

Resources remain fully published and indexed in GBIF even if no specific secondary network is assigned.

IPT Workflow · Step 8: Resource Managers

Role-based access control, collaborative dataset curation, and institutional longevity.



Resource Managers: Role-based access control and collaborative curation

👥 Collaborative Governance

Allows primary authors and data curators (e.g. *Endre Grüner Ofstad* and *Kjetil Pettersson*) to co-manage the dataset.

🔐 Role-Based Access Control

Differentiates between **Resource Creator**, **Manager** (editing mappings/metadata), and institutional **Admins**.

🏛️ Institutional Continuity

Prevents orphaned datasets when individual researchers move institutions — multiple curators ensure long-term maintenance.



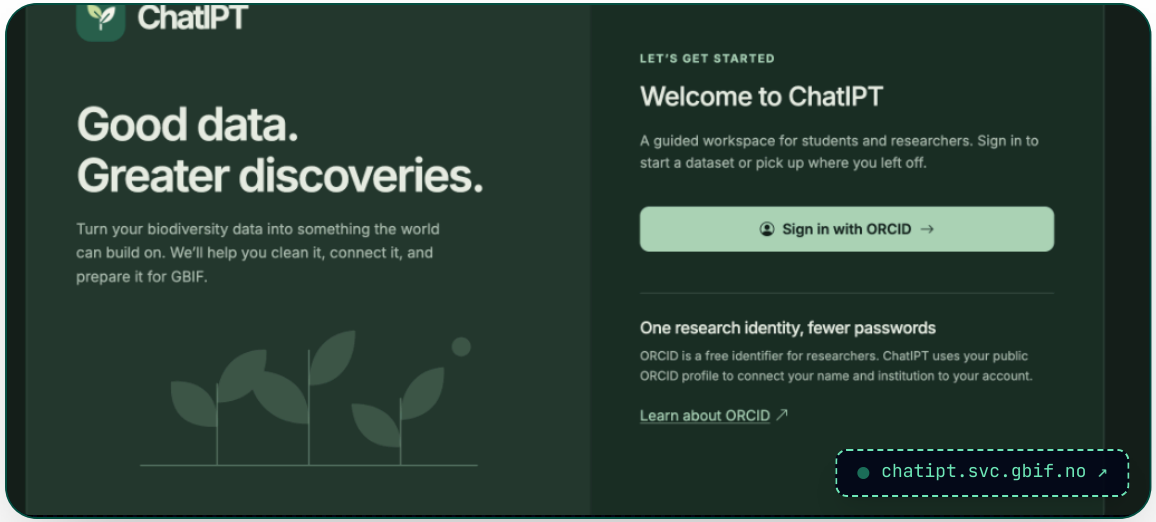
"OK, but that still sounds like a lot of work..."

...Can AI help me somehow?

- Yes! From Darwin Core term mapping to drafting EML metadata and validating records.

Meet ChatIPT: AI-Guided Biodiversity Publishing

A modern, guided AI workspace designed for students and researchers to clean, connect, and prepare datasets for GBIF · chatipt.svc.gbif.no ↗



🌱 Good Data, Greater Discoveries

Guided workspace turning raw spreadsheets into standard, connected biodiversity data.

🤖 AI-Assisted Cleaning & Mapping

Auto-detects column types, recommends Darwin Core terms, and flags spatial errors.

🪪 ORCID-First Authentication

Instant ORCID sign-in tying researcher identity, provenance, and citations directly to datasets.

Launch ChatIPT chatipt.svc.gbif.no ↗

Hands-On Lab: Let’s Try It!

Put biodiversity data publishing into practice — choose your track:

CLASSIC WORKFLOW

GBIF IPT (Sandbox)

The institutional workhorse hosted by national participant nodes for comprehensive dataset administration.

- ✓ Connect raw CSV, TSV, or live SQL databases
- ✓ Map columns directly to Darwin Core Core & Extensions
- ✓ Fill rich EML metadata & manage institutional roles

Recommended for data managers

Open IPT ↗

AI-GUIDED WORKFLOW

ChatIPT Workspace

Intelligent assistant turning raw researcher spreadsheets into validated, connected GBIF datasets.

- ✓ 1-click ORCID authentication (no passwords)
- ✓ AI automated column detection & DwC term suggestions
- ✓ Interactive coordinate cleaning & instant package generation

Recommended for students & researchers

Launch ChatIPT ↗

🎯 **Hands-on Goal:** Take your own sample spreadsheet (or our provided demo CSV), standardize it, and create your first valid Darwin Core dataset!

Hands-on Lab · Lecture 01